# Markov Matrix Formulations for Quasi-Polynomial Dynamics Over Residue Classes

## Introduction

The intersection of probabilistic modeling, ergodic theory, and algebraic number theory frequently gives rise to dynamical systems where continuous phenomena or infinite discrete domains are mapped onto finite topological spaces. Among the most mathematically rich structures in this domain are quasi-polynomials. These functions, which behave globally as polynomials but possess coefficients that are periodic functions of the input, are ubiquitous in advanced mathematics. They originate predominantly from Ehrhart theory, the analysis of vector partition functions, and the formal study of generalized Syracuse (Collatz-type) arithmetic algorithms.[1] Quasi-polynomials inherently exhibit a modular dependence that dictates their trajectories, rendering the long-term behavior of iterated quasi-polynomial maps over the integers notoriously difficult to predict, often brushing against the boundaries of undecidability.

However, by projecting these infinite-domain dynamics onto modular arithmetic spaces—specifically, residue classes modulo a suitably chosen integer $M$ —the evolution of the system can be perfectly modeled as a finite-state Markov chain. This paradigm shift from deterministic, seemingly chaotic number-theoretic functions into stochastic Markov matrix formulations allows for the rigorous deployment of measure theory, spectral analysis, and algebraic geometry to characterize the asymptotic behavior of the system. By defining the states of the Markov chain as the residue classes modulo $M$ , the transition probabilities quantify the exact measure of trajectories flowing from one congruence class to another.[4]

The mathematically rigorous formulation of this transition matrix requires highly specific algebraic and number-theoretic constraints to ensure that the resulting Markov chain is well-behaved. Specifically, the utility of the Markov model hinges upon the transition matrix being both irreducible and aperiodic.[6] When these two fundamental conditions are satisfied, the Perron-Frobenius theorem guarantees the existence of a unique, strictly positive stationary distribution. This distribution offers profound second-order insights into the asymptotic density of the quasi-polynomial orbits, allowing mathematicians to circumvent the intractable nature of individual trajectories by studying the aggregate probabilistic flow of the system.[9]

This exhaustive research report delivers a precise mathematical formulation of finite-state Markov chains derived from a chosen class of well-behaved quasi-polynomials. It details the algebraic, geometric, and number-theoretic conditions required to guarantee the irreducibility

and aperiodicity of the associated transition matrices. Furthermore, it explores the deep implications of these ergodic properties on spectral gaps, mixing times, and the combinatorial geometry of the underlying discrete structures, integrating contemporary research from polyhedral geometry, phylogenetics, and theoretical computer science.

# Algebraic and Analytic Foundations of Quasi-Polynomials

To construct a rigorous Markovian framework, one must first delineate the algebraic structure of the quasi-polynomials under consideration. A function $f : \mathbb{Z} \to \mathbb{Z}$ (or $\mathbb{C}$) is defined as a quasi-polynomial of degree $k$ if there exists a positive integer $d$, termed the quasi-period, and a set of standard polynomials $P_0, P_1, \ldots, P_{d-1} \in \mathbb{Q}[x]$ such that for all integers $n$:

$$f(n) = P_i(n) \quad \text{if } n \equiv i \pmod{d}$$

Equivalently, a quasi-polynomial can be expressed in the standard polynomial form:

$$f(n) = c_k(n)n^k + c_{k-1}(n)n^{k-1} + \cdots + c_0(n)$$

where each coefficient $c_j(n)$ is a periodic function of $n$ with an integral period that divides $d$.[2]

The structural beauty of quasi-polynomials lies in their dual nature. Locally, within a specific residue class modulo $d$, they exhibit smooth, predictable polynomial growth. Globally, the periodic switching between different polynomial branches induces a highly complex, interwoven dynamical system.

## Ehrhart Theory and Vector Partition Functions

Beyond purely arithmetic constructions, quasi-polynomials arise fundamentally in polyhedral geometry through Ehrhart theory.[1] The Ehrhart function $L_P(t)$, which counts the discrete number of integer lattice points inside a polytope $P$ dilated by an integer factor $t$, behaves as a polynomial if the polytope $P$ has integral vertices. However, if the polytope possesses rational vertices, the counting function becomes an Ehrhart quasi-polynomial.[1]

$$L_P(t) = |tP \cap \mathbb{Z}^n| = c_k(t)t^k + \cdots + c_0(t)$$

In this geometric context, the quasi-period $d$ is determined by the lowest common multiple of the denominators of the coordinates of the vertices of the polytope. Vector partition functions, which count the number of ways to express a vector as a non-negative integer linear combination of a fixed set of vectors, also inherently exhibit this quasi-polynomial behavior. These functions describe the number of lattice points in parametric polyhedra where the facets move inward and outward based on input parameters, but the normal vectors to the facets remain constant.[1]

In stochastic processes, such as continuous-time Markov jump-diffusions that preserve polynomial spaces (often utilized in financial modeling or population genetics), the moments of the transition distributions themselves behave as polynomials or quasi-polynomials evaluated at the initial state.[5] Matching these infinite-dimensional moments to finite-state Markov chains represents a profound intersection of measure theory and algebraic combinatorics, necessitating a deep understanding of the underlying matrix formulations.[5]

## The Chosen Class: Generalized Affine Quasi-Polynomial Maps

While the general formulation of quasi-polynomials accommodates arbitrarily high degrees, the mapping of residue classes to finite-state Markov chains is most perfectly realized, and most analytically tractable, within the class of generalized affine quasi-polynomials. In this remarkably well-behaved class, the degree of the constituent polynomials is strictly restricted to $k = 1$, yielding functions of the form:

$$f(n) = \frac{m_i n + r_i}{d} \quad \text{for } n \equiv i \pmod{d}$$

where $m_i$ and $r_i$ are integers chosen such that the condition $m_i n + r_i \equiv 0 \pmod{d}$ is universally satisfied for all $n \equiv i \pmod{d}$, thereby ensuring that $f(n)$ rigidly maps integers to integers.[2]

This specific affine class serves as the mathematical foundation for the generalized Syracuse problem, famously encapsulated by the $3x + 1$ Collatz conjecture. In these models, pioneered by Keith Matthews and other number theorists, the divergent or convergent trajectories of integers are analyzed strictly through their pseudo-random transitions between residue classes.[2]

By restricting the analysis to this affine class, the reduction to a finite state space becomes exact rather than an approximation. The inherent periodicity embedded in the coefficients interacts symmetrically with the modular arithmetic of the constructed state space. This interaction results in transition probabilities that depend exclusively on the current residue class, entirely satisfying the Markov property (memorylessness) and allowing for a perfect

finite-state representation.[18]

# Constructing the Finite-State Markov Matrix Formulation

To mathematically model the transitions between residue classes, one must construct a rigorously defined probability space based on the asymptotic density of integers. Let the finite state space $\Omega$ be defined as the complete set of residue classes modulo a strategically chosen integer $M$.

Often, $M$ is chosen to be the quasi-period $d$ itself. However, to capture higher-order sequential dependencies and resolve hidden Markov states into a fully observable finite-state automaton, $M$ is frequently selected as a higher power, $M = d^k$, where $k \geq 1$.[2] Thus, the state space is formally defined as the cyclic group $\Omega = \mathbb{Z}/M\mathbb{Z} = \{0, 1, \ldots, M-1\}$.

## Derivation of the Transition Matrix and Probability Measure

For a purely deterministic quasi-polynomial $f(n)$, the sequence of iterated compositions $n, f(n), f(f(n)), \ldots$ generates a unique forward orbit for any initial integer $n$. We are fundamentally interested in the transition of a generic integer from a state $i \in \Omega$ to a state $j \in \Omega$ under the application of the map $f$. Since $f$ is deterministic, if the chosen modulus $M$ perfectly divides the quasi-period $d$, the transition from state $i$ to state $j$ is merely a directed, unweighted edge in a deterministic transition graph.[18]

However, a robust Markov matrix formulation must incorporate stochasticity to model the aggregate flow of the entire integer domain. This stochasticity arises naturally and rigorously in two primary mathematical contexts:

First, through the application of the uniform measure over the ring of $p$-adic integers. As established in the Matthews models for generalized Collatz mappings, if one considers an initial seed $n$ chosen uniformly at random from a sufficiently large interval of integers (or, more rigorously, utilizing the normalized Haar measure on the compact topological ring of $p$-adic integers $\mathbb{Z}_p$), the underlying deterministic map induces a precise probability measure on the

transitions between residue classes modulo $M = d^k$.[2] Because the map $f(n)$ divides by $d$, an input known only modulo $M = d^k$ produces an output known only modulo $d^{k-1}$. Thus, to maintain a consistent state space $\Omega$, we assume the "lost" base-$d$ digit of the output is distributed uniformly, generating exact transition probabilities.

Second, stochasticity emerges in randomized quasi-polynomial families. In advanced computational techniques such as simulated annealing, quantum metropolis sampling, or randomized Ehrhart volume estimation via the simple fiber walk, the transition at each discrete time step relies on a randomized choice among a finite family of quasi-polynomial operations.[12]

Let $P$ be the $M \times M$ transition matrix where the matrix element $P_{i,j}$ rigorously denotes the probability of transitioning from residue class $i$ to residue class $j$. Using asymptotic density, this is defined as:

$$P_{i,j} = \lim_{N\to\infty} \frac{|\{n \le N : n \equiv i \pmod{M} \text{ and } f(n) \equiv j \pmod{M}\}|}{|\{n \le N : n \equiv i \pmod{M}\}|}$$

For the generalized affine class $f(n) = \frac{m_i n + r_i}{d}$ evaluated over the modulus $M = d^k$, the transition matrix elements become explicitly computable rational probabilities. These elements form a non-negative, stationary stochastic matrix where the sum of probabilities across any row strictly equals one, ensuring no density escapes the topological space [18]:

$$\sum_{j=0}^{M-1} P_{i,j} = 1 \quad \text{for all } i \in \Omega$$

## Structural Matrix Parameters

The specific algebraic parameters of the quasi-polynomial directly map to the geometric dimensions and sparsity of the resulting Markov transition matrix. The table below details these critical relationships.

| **Quasi-Polynomial Characteristic** | **Implication on Markov Chain Formulation** | **Mathematical Consequence** |
|---|---|---|
| **Quasi-Period ($d$)** | Defines the minimal foundational state space | Transition operations strictly depend only on the |

| | | |
|---|---|---|
| | $\Omega_0 = \mathbb{Z}/d\mathbb{Z}$. | congruence $n \pmod{d}$. |
| **Modulus of Projection ( $M = d^k$ )** | Expands the state space to the ring $\Omega_k = \mathbb{Z}/d^k\mathbb{Z}$. | Captures $k$-step memory, rendering non-Markovian macroscopic behavior into a perfectly memoryless finite-state chain. |
| **Coefficient Multipliers ( $m_i$ )** | Determines the topological transition dispersion (out-degree). | Higher variance in $m_i$ across states leads to denser, more highly connected transition matrices. |
| **Affine Shifts ( $r_i$ )** | Defines the specific target node $j \in \Omega$ for deterministic edges. | Crucial for shifting orbits to avoid absorbing states and ensuring global graph connectivity (irreducibility). |

## Rigorous Conditions for Matrix Irreducibility

A finite-state Markov chain is mathematically classified as irreducible if it is structurally possible to reach any target state from any initial state in a finite number of discrete time steps.[6]

Formally, for all pairs of states $i, j \in \Omega$, there must exist some integer $n \geq 1$ such that the $n$-step transition probability—represented by the matrix power—is strictly positive: $(P^n)_{i,j} > 0$.[7]

In the specific context of quasi-polynomials modeled over residue classes, ensuring irreducibility translates to finding the precise number-theoretic and algebraic conditions under which the directed transition graph induced by the matrix $P$ is strongly connected. If the underlying graph contains disjoint subgraphs, the state space irrevocably fractures into isolated communicating classes. This violates irreducibility, negates the Ergodic theorem, and leads to a

system with multiple divergent stationary distributions heavily dependent on the initial seed.[23]

## First Condition: Coprimality and Density of Multipliers

For the generalized affine map $f(n) = \frac{m_i n + r_i}{d}$ defined on residue classes modulo $d$, irreducibility is overwhelmingly dictated by the prime factorization of the multipliers $m_i$ relative to the modulus $d$. If there exists a common arithmetic divisor $g = \gcd(m_0, m_1, \ldots, m_{d-1}, d) > 1$, the state space immediately and necessarily partitions into distinct modular orbits that can never intersect.

To guarantee that a transition matrix is irreducible, the generating quasi-polynomial map must be of a "relatively prime type".[2] Specifically, the transition matrix $P$ is irreducible if and only if the algebraic ideal generated by the set of all multipliers and the modulus spans the entire integer ring $\mathbb{Z}$. Furthermore, to prevent the chain from collapsing into a lower-dimensional linear subspace of residue classes, it is geometrically necessary that for every prime $p$ dividing the modulus $M$, the mapping does not globally preserve congruences modulo $p$. This ensures that the probabilistic flow diffuses across the entire modular topology.

## Second Condition: Absence of Trapping Cycles and Absorbing States

An absorbing state $i$ in a Markov chain is defined as a state where the self-transition probability is absolute, $P_{i,i} = 1$, making it topologically impossible for the process to leave once the state is entered.[23] For deterministic quasi-polynomials, an absorbing state modulo $M$ corresponds to a residue class $i$ such that $f(i) \equiv i \pmod{M}$ and no other probabilistic transitions out of $i$ are permissible.

A sufficient mathematical condition to guarantee the absence of absorbing states—and more generally, to guarantee the absence of closed, non-spanning communicating classes—is the existence of a balanced set of roots within the split polynomial representation of the transition mapping.[26] If the distribution of elements among the residue classes of powers of $M$ is combinatorially balanced, no single congruence class or small cycle of classes can act as an infinite, inescapable sink. The flow of trajectories is forced to cascade through the entirety of $\Omega$.

## Third Condition: Uniform Non-Zero Sub-Stochastic Deficit (Regular

### Splittings)

In theoretical scenarios dealing with sub-stochastic matrices—where certain rows sum to strictly less than one due to trajectories escaping an artificially restricted analytic boundary—irreducibility on the restricted space requires continuous internal cyclic flow. By employing non-negative matrix theory, specifically M-matrices and their regular splittings, one can analyze the spectral radius of the transition components.[10]

If the generalized quasi-polynomial is artificially adjusted via PageRank-style damping or teleportation factors (incorporating an external uniform distribution across all congruence classes), irreducibility is structurally guaranteed because every matrix element receives a strictly positive additive term.[27] However, for the pure, unadulterated quasi-polynomial transition matrix, the rigorously required condition is that the union of the ranges of the polynomial branches $P_i(x) \pmod{M}$ completely, uniformly covers the ring $\mathbb{Z}/M\mathbb{Z}$.

In the advanced language of vector partition functions and Ehrhart polynomials, the irreducibility of the random walk on the affine semigroup is strictly equivalent to the geometric requirement that the generating rays of the parametric polyhedron rationally span the associated vector space.[12] If the defining rays fall into a lower-dimensional hyperplane, the Markov chain executed on the residue classes modulo the determinant of the underlying lattice will be inherently reducible, trapping the stochastic process in a subspace of the polytope's interior.

## Rigorous Conditions for Matrix Aperiodicity

A specific state $i$ in a Markov chain is defined as aperiodic if its period—defined formally as the greatest common divisor (gcd) of all discrete time steps $n \geq 1$ such that the return probability $(P^n)_{i,i} > 0$—is exactly 1.[6] By extension, a Markov chain is globally defined as an aperiodic chain if all its constituent states are aperiodic.[7]

Aperiodicity is a foundational requirement for the strict application of the Ergodic Theorem. The theorem states that an irreducible, aperiodic Markov chain with a finite state space possesses a unique stationary distribution $\pi$ to which the sequence of distributions at time $t$ converges smoothly from any arbitrary initial distribution as time $t \to \infty$.[6] Without aperiodicity, the chain may oscillate deterministically between subsets of the state space, preventing the distribution from ever settling into a stationary equilibrium.

### First Condition: Existence of a Coprime Cycle Basis

To guarantee that the transition matrix associated with a quasi-polynomial over $\mathbb{Z}/M\mathbb{Z}$ is

aperiodic, it is mathematically sufficient to prove the existence of at least two topological cycles within the state transition graph whose lengths are relatively prime to one another.

Let $\mathcal{C}$ denote the mathematical set of all possible cycle lengths in the directed transition graph of the quasi-polynomial modulo $M$. The Markov chain is aperiodic if and only if $\gcd(\mathcal{C}) = 1$. In the specific context of quasi-polynomials, which map $x \mapsto P_i(x) \pmod{M}$, cycles correspond directly to modular periodic orbits. By rigorously analyzing the degree of the constituent polynomials, one can invoke the Weil conjectures—specifically via the Hasse-Weil bound for algebraic curves over finite fields—to estimate the number of periodic points. For a sufficiently well-behaved quasi-polynomial mapped over a large modulus $M$ (especially when $M$ is prime), the existence of both a fixed point (a cycle of length 1) and higher-order multi-step cycles naturally occurs, satisfying the condition $\gcd(\mathcal{C}) = 1$ and preventing rhythmic oscillations.

## Second Condition: The Fixed-Point Guarantee (Self-Loops)

The most direct and computationally verifiable mechanism to guarantee aperiodicity in a known irreducible Markov chain is to establish the existence of at least one self-loop in the transition graph.[6] A self-loop exists if there is some state $i$ such that the immediate return probability is positive: $P_{i,i} > 0$. It is a fundamental theorem of stochastic processes that if an irreducible chain possesses even a single state with a period of 1, the period of the entire chain collapses, rendering all states aperiodic.[7]

For the quasi-polynomial $f(n)$, a self-loop exists if there is at least one residue class $i \in \{0, \ldots, M-1\}$ such that:

$$f(i) \equiv i \pmod{M}$$

For the affine class $f(n) = \frac{m_i n + r_i}{d}$, evaluating this structural requirement modulo $M$ necessitates finding an integer $i$ such that:

$$m_i i + r_i \equiv d \cdot i \pmod{M \cdot d}$$

This translates immediately into a standard linear congruence problem. If the greatest common divisor $\gcd(m_i - d, M \cdot d)$ evenly divides the shift parameter $r_i$ for at least one branch of

the quasi-polynomial, then an exact fixed point exists modulo $M$. Consequently, the transition matrix $P$ will possess a strictly non-zero element on its principal diagonal, categorically guaranteeing the aperiodicity of the Markov chain.[6]

### Third Condition: Lazification of the Chain

In practical cases where the natural arithmetic dynamics of the quasi-polynomial inherently force bipartite or $k$-partite deterministic hopping between residue classes—such as sequences that strictly alternate between even and odd outputs at every step—the underlying graph is fundamentally periodic.

To systematically overcome this structural limitation, a standard probabilist's technique known as *lazification* is applied. By modifying the transition matrix to $P' = \frac{I+P}{2}$, where $I$ is the identity matrix, the modified chain holds its current state with a probability of $1/2$ at each discrete step.[29] This profound mathematical adjustment preserves both the original stationary distribution $\pi$ and the irreducibility of the graph, while artificially introducing robust self-loops at every single state. This instantly renders the modified transition matrix $P'$ strongly aperiodic.[24] This technique is extensively and crucially utilized when modeling Ehrhart quasi-polynomial fluctuations via Markov Chain Monte Carlo (MCMC) algorithms, where discrete parity phenomena in the lattice points would otherwise categorically prevent algorithmic convergence.[12]

## Spectral Geometry, Ergodicity, and Mixing Time Bounds

Once the algebraic parameters of the quasi-polynomial have been meticulously tuned to guarantee an irreducible and aperiodic transition matrix $P$, the analytic focus shifts into the domain of Perron-Frobenius theory, non-negative matrices, and spectral geometry. The transition matrix is an $M \times M$ non-negative real matrix.[27]

### The Perron-Frobenius Theorem and Stationary Distributions

By the Perron-Frobenius Theorem, an irreducible and aperiodic non-negative matrix $P$ possesses a unique real eigenvalue of largest magnitude, which for a stochastic matrix is strictly $\lambda_1 = 1$. Furthermore, its corresponding left eigenvector, denoted $\pi$, is unique up to scaling and strictly contains positive entries across all its coordinates.[6] By normalizing this

vector such that its elements sum to 1, it constitutes the unique, invariant stationary distribution of the Markov chain:

$$\pi P = \pi, \quad \sum_{i=0}^{M-1} \pi_i = 1$$

The vector $\pi$ describes the exact asymptotic fraction of time the quasi-polynomial orbit spends in each residue class modulo $M$. For generalized Collatz mappings, determining whether $\pi$ is uniform across all states or heavily skewed toward specific residue classes provides the absolute foundation for predicting whether random trajectories diverge to infinity or collapse into known bounded cycles.[2]

## The Spectral Gap, Conductance, and Mixing Times

The exponential rate at which the distribution of the state variable $X_t$ (the residue class at time $t$) converges to the invariant stationary distribution $\pi$ is governed tightly by the Second Largest Eigenvalue Modulus (SLEM) of the matrix $P$. This is denoted as $\lambda_* = \max_{j>1} |\lambda_j|$.[31] The absolute spectral gap of the transition matrix is defined as $\gamma = 1 - \lambda_*$.

The total variation distance between the state distribution after $t$ transitions, $P^t(x, \cdot)$, and the ultimate stationary distribution $\pi$ is strictly bounded by the spectral gap [21]:

$$\max_{x \in \Omega} \|P^t(x, \cdot) - \pi\|_{TV} \leq C\lambda_*^t = C(1-\gamma)^t$$

For quasi-polynomials arising from random walks on affine semigroups (intimately related to Ehrhart polynomials), analytically bounding the spectral gap is a highly complex topological task. The mixing time $t_{mix}$ typically scales polynomially or quasi-polynomially with the dimension of the underlying polytope.[12] If the quasi-polynomials represent counting functions for lattice points, the bottleneck ratio, also known as the conductance, denoted $\Phi^*$, offers a vital geometric method to bound the SLEM [31]:

$$\Phi^* := \min_{A \subset \Omega, \pi(A) \leq \frac{1}{2}} \frac{\sum_{i \in A, j \in A^c} \pi_i P_{i,j}}{\pi(A)}$$

By Cheeger's celebrated inequality, the spectral gap is constrained by the geometry of the state space: $\frac{(\Phi^*)^2}{2} \leq \gamma \leq 2\Phi^*$. In modular transition matrices strictly derived from well-behaved affine quasi-polynomials, the conductance is heavily influenced by the presence

of dense arithmetic progressions embedded within the constituent maps $P_i(n)$. If the multipliers $m_i$ are small relative to the modulus $M$, the state space exhibits severe geometric bottlenecks. This restriction leads to exponentially slow, quasi-polynomial time mixing, represented analytically by bounds of the form $\lambda_* \geq 1 - 2\left[\exp\left(\frac{2\log(C_1 n)}{C_0 \cdot i} - 1\right)\right]$.[31]

# Advanced Case Studies and Phenomenological Applications

The theoretical framework established above is not merely an abstract mathematical exercise; it represents the primary functional engine for solving complex problems across multiple branches of advanced mathematics.

## 1. Matthews' Matrix for the Generalized Syracuse Problem

The most direct and historically rich application of this specific Markovian theory is seen in Keith Matthews' profound investigations of generalized Syracuse algorithms, which extend the famous $3x + 1$ Collatz conjecture.[2] Let $d$ be a chosen modulus, and define a comprehensive set of affine maps $f_i(x) = \frac{m_i x - r_i}{d}$ for any integer $x \equiv i \pmod{d}$.

Matthews meticulously constructs a transition matrix $P$ describing the progression of trajectory densities through the residue classes modulo $M = d^k$. By proving that the formulated matrix $P$ is irreducible and aperiodic (which relies heavily on the relative primality of the multipliers $m_i$ and the modulus $d$), one establishes that the proportion of trajectory steps spent in any specific residue class converges precisely to the scalar components of the stationary vector $\pi$.[2]

If the stationary distribution leans heavily toward residue classes that trigger multiplication by a factor $m_i > d$, the trajectory is biased toward inflation and divergence to infinity. Conversely, if $\pi$ weights the transitions toward states where $m_i < d$, the trajectories generally decay and collapse into tight periodic cycles.[2] This transition matrix modeling successfully reduces a notoriously intractable, seemingly undecidable number theory problem into a highly evaluable linear algebra eigenvalue calculation. The accuracy of these quasi-polynomial Markov models even led to a famous $100 AUD prize offered by Matthews for rigorous proofs regarding the eventual convergence of specific divergent trajectories within these congruence classes.[17]

## 2. Ehrhart Quasi-Polynomials and Polyhedral MCMC Sampling

In the field of polyhedral geometry, the exact number of integer points contained in a parameterized rational polytope $P$ dilated by a factor $t$ is given by an Ehrhart quasi-polynomial $L_P(t)$.[1]

$$L_P(t) = c_r(t)t^r + c_{r-1}(t)t^{r-1} + \cdots + c_0(t)$$

where $c_i(t)$ are periodic functions in $t$ with periods strictly dividing the lowest common multiple of the denominators of the rational vertices of $P$.[3] To computationally estimate the vast volume of these high-dimensional polytopes, algorithms deploy simple fiber walks, hit-and-run sampling, or sophisticated Markov Chain Monte Carlo (MCMC) algorithms.

The probabilistic transitions between the internal discrete states of these semigroups form a finite-state Markov chain when reduced modulo the quasi-period. The mathematical irreducibility of this chain guarantees that the random walk explores the entirety of the polytope's volume without becoming trapped. Meanwhile, aperiodicity (often artificially introduced via lazification to ensure spectral independence [29]) ensures that the sampler achieves a stationary distribution directly proportional to the true volume of the target region.[12] The failure to achieve an irreducible transition matrix in these settings results in the sampler becoming isolated within a subset of the polytope, leading to catastrophic, algorithm-breaking underestimations of the Ehrhart coefficients.[12] Current research heavily utilizes these quasi-polynomial properties to achieve fully polynomial-time randomized approximation schemes (FPRAS), avoiding the quasi-polynomial time barriers inherent to unoptimized geometric samplers.[32]

## 3. Moment Matching in Polynomial-Preserving Processes

In systems where quasi-polynomials preserve polynomial state spaces—such as continuous-time Markov jump-diffusions extensively utilized in quantitative financial modeling and phylogenetic tree construction—approximating the continuous random process via a finite-state Markov chain introduces moments that are themselves pure polynomials in the initial state.[5]

The fundamental characterization theorem for such advanced processes indicates that constructing an exact finite-state Markov process that flawlessly matches moments of arbitrary high order is generally impossible for polynomial diffusions in continuous time.[5] However, a discrete-time construction utilizing robust cubature methods over long-run moments makes the finite-state Markov modeling entirely feasible. Here, the carefully verified irreducible and aperiodic properties of the constructed residue class matrix guarantee the

existence of these long-run moments. This algebraic stability allows the continuous quasi-polynomial volume and trajectory to be captured perfectly by the discrete finite-state approximations.[5]

## Synthesis: The Translation of Algebraic Geography to Probabilistic Flow

A profound second-order mathematical insight arises when deeply observing how the periodic coefficients of the quasi-polynomial perfectly dictate the topological shape of the transition graph. It is not merely that random transitions occur; rather, the arithmetic properties of the quasi-polynomials—specifically the prime congruences of the multipliers—act as precise mathematical "valves" that direct the probabilistic flow through the state space.

When transitioning continuously between residue classes, the quasi-polynomial maps essentially perform an ergodic scrambling of the integer number line. If the transition matrix $P$ is proven to be irreducible, it categorically implies that the quasi-polynomial possesses no localized arithmetic traps. It mathematically suggests that any integer sequence subjected to these operations will eventually, and uniformly, "forget" its deterministic starting value concerning its modulo $M$ representation.

This translates directly to a vital third-order insight: The unpredictability, or pseudo-randomness, of integer sequences generated by these mappings is absolutely not an artifact of random choice. Instead, it is an emergent, deterministic property of the periodic dissonance between the quasi-period $d$ and the multipliers $m_i$. By formulating this entire dynamical system perfectly as a finite-state Markov chain, one essentially bypasses the necessity of tracking exact, exploding integer coordinates. The analyst opts instead to analyze the continuous probabilistic density of the orbits.

This specific paradigm shift proves vital when attempting to resolve extreme undecidability problems in discrete mathematics. While it may be fundamentally impossible to determine if a specific integer $n$ reaches $1$ (as in the traditional $3x+1$ problem), rigorously modeling the system as an aperiodic, irreducible Markov chain allows mathematicians to prove that *almost all* integer trajectories adhere strictly to a determinable, global probability distribution across residue classes.[2] Furthermore, observing whether these matrices satisfy detailed balance (reversibility) provides an informational arrow of time; the absence of reversibility in affine quasi-polynomials highlights the asymmetric entropy generation inherent to generalized Syracuse processes.[22]

## Comprehensive Matrix Framework and Criteria Matrix

To systematically guarantee that the transition matrix $P$ generated by a chosen quasi-polynomial $f(n)$ over the ring $\mathbb{Z}/M\mathbb{Z}$ behaves perfectly as an ergodic finite-state Markov chain, the following conditions must be concurrently and strictly satisfied. The table below consolidates the theoretical framework.

| **Matrix Characteristic** | **Algebraic Requirement on f(n)=Pi(n)(modd)** | **Topological Outcome on Markov Graph** |
|---|---|---|
| **Row Stochasticity** | $\sum_{j=0}^{M-1} P_{i,j} = 1 \quad \forall i \in \Omega$ | Transitions map completely into $\mathbb{Z}/M\mathbb{Z}$ with absolutely no escaping density. |
| **Irreducibility** | $\gcd(m_0, m_1, \ldots, m_{d-1}, M$ | The graph is strongly connected; there are no isolated subgraphs or invariant subspaces. |
| **Aperiodicity** | $\exists i \in \mathbb{Z}/M\mathbb{Z}$ such that $f(i) \equiv i \pmod{M}$ | The greatest common divisor of all cycle lengths is strictly 1, averting deterministic periodic oscillation. |
| **Reversibility** | $\pi_i P_{i,j} = \pi_j P_{j,i}$ (Requires Detailed Balance) | Symmetric probabilistic flow between congruence classes, allowing Hermitian spectral analysis. |
| **Rapid Mixing** | Spectral Gap $\gamma \geq \Omega(M^{-c})$ for constant $c$ | Total variation distance from the stationary state decays exponentially, avoiding quasi-polynomial time |

| | | |
|---|---|---|
| | | algorithmic blockages. |

# Conclusion

The algebraic projection of well-behaved quasi-polynomial functions onto finite residue classes provides an immensely powerful mathematical framework for rendering intractable arithmetic systems into solvable linear algebra and probability problems. By precisely defining the states of a Markov chain as the complete set of congruence classes modulo a scaling factor $M$, the infinitely expansive behavior of integer orbits is mapped directly, and perfectly, into a finite $M \times M$ probability transition matrix.

For this stochastic model to function with mathematical perfection—ensuring the absolute convergence of state distributions over time—the formulated matrix must be rigorously proven to be both irreducible and aperiodic. This exhaustive report has demonstrated that irreducibility strictly requires the constituent coefficients of the quasi-polynomial to avoid common arithmetic divisors with the defined modulus. This guarantees that the directed graph of modular transitions remains strongly connected without succumbing to disjoint communicating classes or inescapable absorbing sinks. Concurrently, aperiodicity is secured either through the existence of algebraic fixed points modulo $M$, or through the probabilistic technique of matrix lazification, which artificially breaks deterministic, rhythm-locked oscillations without altering the stationary limits.

When these precise algebraic and topological conditions are met, the Perron-Frobenius theorem securely anchors the dynamical system, revealing a unique stationary distribution vector that perfectly reflects the fundamental long-term densities of the quasi-polynomial's underlying integer trajectories. Whether analyzing the ergodic limits of generalized Syracuse arithmetic maps, matching moments in continuous-time polynomial jump-diffusions, or bounding the polynomial mixing times of complex volume samplers over rational polytopes in Ehrhart theory, the finite-state Markov chain formulation stands as an indispensable analytical bridge between discrete algebraic topology and advanced applied stochastic modeling.

### Works cited


1. ABSTRACTS - American Mathematical Society, accessed March 12, 2026, https://www.ams.org/journals/abs/2012-33-04/abs-33-04.pdf?active=allissues
2. Mathematicians' Collatz Conundrum | PDF | Dynamical System | Number Theory - Scribd, accessed March 12, 2026, https://www.scribd.com/document/231794502/3x-1-problem-an-overview-j-lagarias

3. ABSTRACTS - American Mathematical Society, accessed March 12, 2026, https://www.ams.org/journals/abs/2009-30-03/abs-30-03.pdf?active=allissues
4. Learn Quasi-Stationary Distributions of Finite State Markov Chain - PMC, accessed March 12, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8774945/
5. Finite-state approximation of polynomial preserving processes - Mathematical Institute, accessed March 12, 2026, https://www.maths.ox.ac.uk/node/10958
6. Characterizing the Aperiodicity of Irreducible Markov Chains by Using P Systems - idUS, accessed March 12, 2026, https://idus.us.es/items/a8d3c017-a09f-46b0-ab11-1ce9bacdb230
7. Irreducible and Aperiodic Markov Chains, accessed March 12, 2026, https://www.mathematik.uni-ulm.de/stochastik/lehre/ss06/markov/skript_engl/node12.html
8. Lecture Markov chains properties: irreducibility and aperiodicity 1e4etiti:n The defintion fix three things, accessed March 12, 2026, https://pages.dataiku.com/hubfs/Dataiku%20Dec%202016/Files/lecture3.pdf
9. markov.pdf - Université de Rennes, accessed March 12, 2026, https://perso.univ-rennes1.fr/dimitri.petritis/enseignement/markov/markov.pdf
10. A family of fast fixed point iterations for M/G/1-type Markov chains† - Dipartimento di Matematica, accessed March 12, 2026, https://people.dm.unipi.it/bini/Bini-Latouche-Meini_IMA.pdf
11. Shortest Paths, Markov Chains, Matrix Scaling and Beyond: Improved Algorithms Through the Lens of Continuous Optimization - DSpace@MIT, accessed March 12, 2026, https://dspace.mit.edu/bitstream/handle/1721.1/112828/1014342852-MIT.pdf?sequence=1
12. Fiber graphs, accessed March 12, 2026, https://opendata.uni-halle.de/bitstream/1981185920/12249/1/Dissertation_Windisch.pdf
13. Enumeration in stochastic processes and polyhedral geometry - DASH, accessed March 12, 2026, https://dash.harvard.edu/bitstreams/e29c3b6a-26b2-4a0a-abbf-a37d46a30995/download
14. Mathematisches Forschungsinstitut Oberwolfach Arrangements, Matroids and Logarithmic Vector Fields - EMS Press, accessed March 12, 2026, https://ems.press/content/serial-article-files/49485
15. MSJ 2024 Autumn Meeting - Titles and Short Summaries of the Talks, accessed March 12, 2026, https://www.mathsoc.jp/en/assets/file/meeting/osaka24sept/summary24sept_en_20240821.pdf
16. Infinite root systems in algebra and geometry & Enumeration in stochastic processes and polyhedral geometry - Harvard Math, accessed March 12, 2026, https://www.math.harvard.edu/event/infinite-root-systems-in-algebra-and-geometry-enumeration-in-stochastic-processes-and-polyhedral-geometry/
17. Collatz 3x+1 Conjecture Proved! -.:: GEOCITIES.ws ::., accessed March 12, 2026, http://www.geocities.ws/collatz_conjecture_proved/submitted_to_ams_bulletin.pd

f
18. Probabilistic Machine Learning: Advanced Topics (Adaptive Computation and Machine Learning series) 0262048434, 9780262048439 - DOKUMEN.PUB, accessed March 12, 2026, https://dokumen.pub/probabilistic-machine-learning-advanced-topics-adaptive-computation-and-machine-learning-series-0262048434-9780262048439.html
19. Probabilistic Machine Learning: Advanced Topics [2, Draft ed.] 9780262048439, 9780262376006, 9780262375993 - DOKUMEN.PUB, accessed March 12, 2026, https://dokumen.pub/probabilistic-machine-learning-advanced-topics-2-draftnbsped-9780262048439-9780262376006-9780262375993.html
20. 1 Probability - Ahmad Humayun, accessed March 12, 2026, https://ahumayun.com/resources/notes/humayun_machine-learning_notes.pdf
21. Rigorous Analysis of Markov Processes with Applications to Quantum Information - UNM Digital Repository, accessed March 12, 2026, https://digitalrepository.unm.edu/cgi/viewcontent.cgi?article=1283&context=phyc_etds
22. An Efficient Randomized Approximation Algorithm for Volume Estimation and Design Centering, accessed March 12, 2026, https://publications.mpi-cbg.de/Asmus_2017_6878.pdf
23. BSc in Actuarial Sciene Syllabuses and Regulations (4-year curriculum) 2019-2020, accessed March 12, 2026, https://www.scifac.hku.hk/f/page/5822/13121/BSc(AC)4_2019-2020.pdf
24. Behavioral Game Theory - ETH Zürich, accessed March 12, 2026, https://ethz.ch/content/dam/ethz/special-interest/gess/computational-social-science-dam/documents/internal/hnax/habil.pdf
25. [2409.19246] Quasi-stationary distributions of non-absorbing Markov chains - arXiv.org, accessed March 12, 2026, https://arxiv.org/abs/2409.19246
26. What You Should Know About Integer-Valued Polynomials | Request PDF - ResearchGate, accessed March 12, 2026, https://www.researchgate.net/publication/301706106_What_You_Should_Know_About_Integer-Valued_Polynomials
27. Perturbed Markov Chains with Damping Component and Information Networks - Diva-Portal.org, accessed March 12, 2026, https://www.diva-portal.org/smash/get/diva2:1476792/FULLTEXT02.pdf
28. (ML 18.3) Stationary distributions, Irreducibility, and Aperiodicity - YouTube, accessed March 12, 2026, https://www.youtube.com/watch?v=tByUQbJdt14
29. Spectral Independence A New Tool to Analyze Markov Chains Kuikui Liu A dissertation submitted in partial fulfillment of the requ, accessed March 12, 2026, https://kuikuiliu.github.io/files/dissertation.pdf
30. Non-negative Matricesand Markov Chains, accessed March 12, 2026, http://www.yaroslavvb.com/papers/seneta-nonnegative1.pdf
31. ABSTRACT - NC State Repository, accessed March 12, 2026, https://repository.lib.ncsu.edu/bitstreams/bd174095-379e-4f38-a235-b65ee4447e0e/download
32. Configuring Random Graph Models with Fixed Degree Sequences | SIAM Review

- Computer Science, accessed March 12, 2026, https://www.cs.cornell.edu/courses/cs6241/2019sp/readings/Fosdick-2018-configuration.pdf
33. Approximate counting, phase transitions and geometry of polynomials - EECS, accessed March 12, 2026, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2019/EECS-2019-110.pdf
34. Lectures on Algebraic Statistics, accessed March 12, 2026, https://math.berkeley.edu/~bernd/owl.pdf